The network types in use

Type	The network emits	End of a step	Loss
1	the stage states	summed from the stages	pooled or cost-weighted
2	the correction to the straight line, scaled per track	summed from the stages	pooled
3	the correction to the straight line, one fixed scale	summed from the stages	pooled

Colours in the schematics: **blue** input, **grey** network, **orange** raw output to state, **green** physics with no trained weights, **red** loss, **yellow** reference states.

1 Common to all types

State and equation of motion

The region is $z_0 = 2648.2\,\text{mm}$ to $z_1 = 7826.0\,\text{mm}$, length $L = 5177.8\,\text{mm}$. The state on a plane of constant z is $S = (x, y, t_x, t_y, q/p)$. The rates are

$$\frac{dx}{dz} = t_x, \qquad \frac{dt_x}{dz} = \kappa\,\frac{q}{p}\sqrt{1 + t_x^2 + t_y^2}\,\big(t_x t_y B_x - (1 + t_x^2)B_y + t_y B_z\big),$$
$$\frac{dy}{dz} = t_y, \qquad \frac{dt_y}{dz} = \kappa\,\frac{q}{p}\sqrt{1 + t_x^2 + t_y^2}\,\big((1 + t_y^2)B_x - t_x t_y B_y - t_x B_z\big),$$

with $d(q/p)/dz = 0$, $\kappa = 10^{-3}$, and B from the field map. q/p is an input to every step and is never predicted. Write $f(S, z)$ for the four rates.

One network, chained

One network serves one step length $\Delta z = L/N$. It is applied N times. Step k starts on $z_k = z_0 + k\Delta z$. The end state of step k, with q/p copied, is the input of step $k+1$. Δz is a constant of the network, not an input.

The Gauss–Legendre step

The network emits the q stages of a Gauss–Legendre step (tableau A, weights b, nodes c). Stage j is the state on the plane $z_k + c_j\Delta z$. The exact stages solve

$$\hat{S}_j = S + \Delta z \sum_{m=1}^{q} A_{jm}\, f(\hat{S}_m,\ z_k + c_m\Delta z), \qquad j = 1, \ldots, q.$$

The end of the step is collected and summed from the stages:

$$S_{\text{end}} = S + \Delta z \sum_{m=1}^{q} b_m\, f(\hat{S}_m,\ z_k + c_m\Delta z).$$

The network body

- 6 inputs: S divided by the spread of each component over the round-1 training states, and z_k mapped linearly onto $[-1, 1]$.
- 2 hidden layers of 128 units, tanh.
- A linear output layer of $4q$ numbers: q stages $\times$ (x, y, t_x, t_y).
- Double precision.

The types differ only in how the $4q$ raw outputs become stage states.

Tracks

Simulated particles, each carried across the region by a sixth-order Runge–Kutta integrator at 0.1 mm steps. The state is stored on 257 planes. This is the reference.

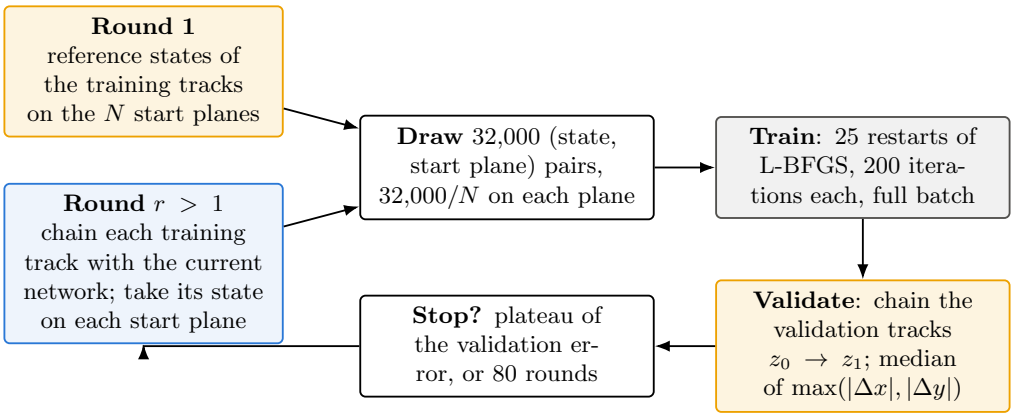
Split	Tracks	Use
training	11,567	round-1 states; start states of the chains in later rounds
validation	1,463	stopping rule
test	1,452	scores

No labels

The loss uses only the equation of motion and the network outputs. No reference state is in the loss. The reference is used for the round-1 training states, for validation, and for test.

2 How the data is fed

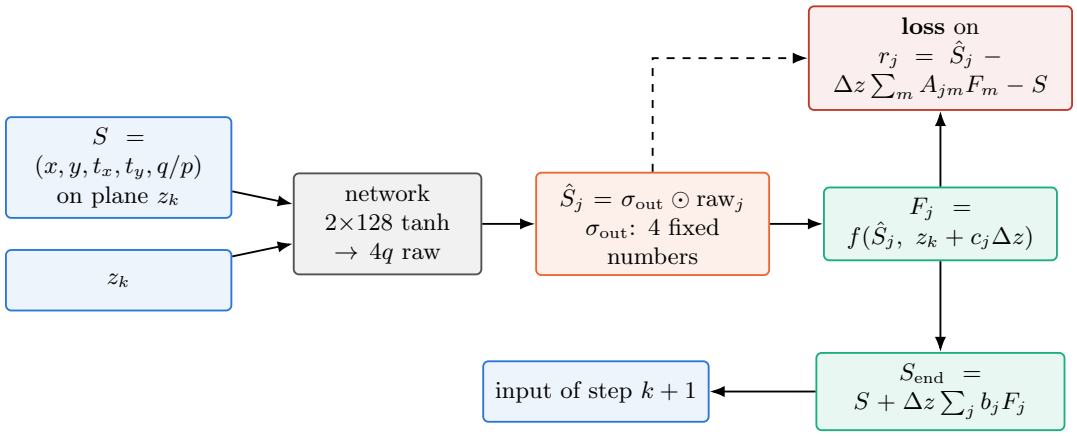
Training is in rounds.



- The 32,000 states of a round are fixed during the round.
- Each state is one step on its own. No gradient flows through the chain.
- From round 2 the network trains on the states its own chain visits.
- The loss is rescaled to 1 at the start of each round, and again with a new optimiser each time it falls by a factor 10.
- Plateau: the median validation error of the last 10 rounds is at least 0.95 of the median of the 10 before, at each of the last 3 rounds.
- The input scales, the output scales of types 1 and 3, and the loss constants are measured once, on the round-1 states.

3 Type 1: direct stage states

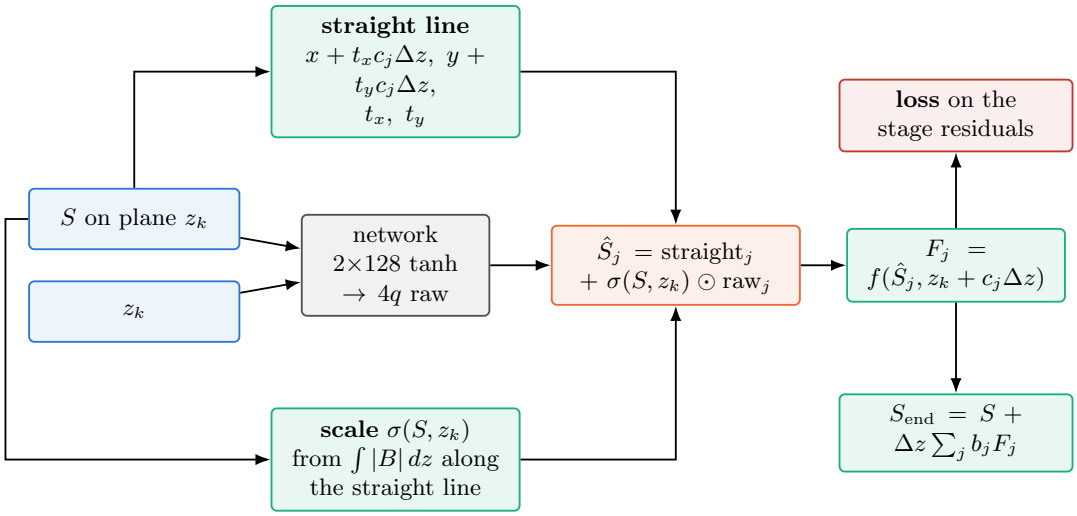
The network emits the stage states. A raw output of zero is the origin of the detector.



Input	S and z_k.
Raw output	$4q$ numbers.
Prediction	The q stage states: $\hat{S}_j = \sigma_{\text{out}} \odot \text{raw}_j$. σ_{out} is the spread of (x, y, t_x, t_y) over the round-1 states (445 mm, 365 mm, 0.164, 0.069 at $N = 64$).
End of step	$S + \Delta z \sum_j b_j f(\hat{S}_j, z_k + c_j \Delta z)$.
Target	None.
Loss	Pooled or cost-weighted, on the q stage residuals.
Configurations	64 steps / 2 stages, 128 / 8, 256 / 16.

4 Type 2: straight line plus a correction, scaled per track

The network emits the correction to the straight line. The correction is multiplied by a scale computed for each track from its state and the field along its straight line. A raw output of zero is the straight line.



The scale. Along the straight line through S over one step, with 16 midpoint samples:

$$I_B = \int |B|\, dz, \qquad I_y = \int \sqrt{1 + t_x^2 + t_y^2}\, \big((1 + t_y^2)|B_x| + |t_x t_y B_y| + |t_x B_z|\big)\, dz,$$

$$\beta_x = \kappa\, |q/p|\, I_B, \qquad \beta_y = \max\big(\kappa\, |q/p|\, I_y,\ 10^{-3}\beta_x\big),$$

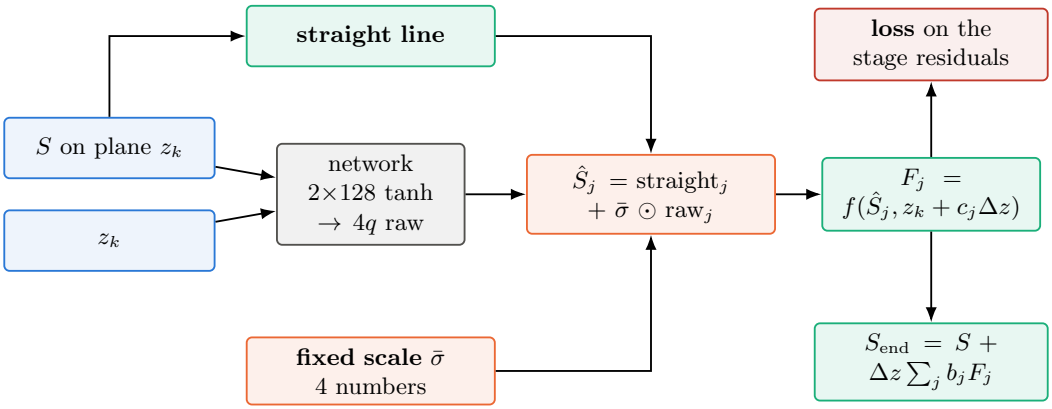
$$\sigma = \big(\beta_x \Delta z/2,\ \beta_y \Delta z/2,\ \beta_x,\ \beta_y\big) \quad \text{for } (x, y, t_x, t_y).$$

β is the size of the change of slope over the step. $\beta\Delta z/2$ is the size of the change of position. The scale has no trained weights.

Input	S and z_k.
Raw output	$4q$ numbers.
Prediction	The correction of each stage to the straight line, in units of σ: $\hat{S}_j = \text{straight}_j(S) + \sigma(S, z_k) \odot \text{raw}_j$.
End of step	$S + \Delta z \sum_j b_j f(\hat{S}_j, z_k + c_j \Delta z)$.
Target	None.
Loss	Pooled, on the q stage residuals.
Configurations	2 stages at 8, 16, 32, 64, 128, 256 steps; 4, 8, 16 stages at 64 steps.

5 Type 3: straight line plus a correction, one fixed scale

As type 2, but the scale is four fixed numbers for all tracks: the median of the per-track scale σ over the round-1 states.



Prediction	The correction to the straight line, in units of $\bar{\sigma}$. At 64 steps $\bar{\sigma}$ = (0.0957 mm, 0.00085 mm, 0.00237, 0.0000209).
Configurations	64 steps / 2 stages.
Everything else	As type 2.

6 Losses

Stage residual. For training state n, stage j, component $d \in \{x, y, t_x, t_y\}$:

$$r_{njd} = \hat{S}_{njd} - \Delta z \sum_{m=1}^{q} A_{jm}\, f_d\big(\hat{S}_{nm},\ z_n + c_m \Delta z\big) - S_{nd}.$$

It is zero for all n, j, d only when the stages solve the Gauss–Legendre equations. It has no label.

Pooled

$$\mathcal{L} = \operatorname*{mean}_{n,j,d} \left(\frac{r_{njd}}{s_d}\right)^2,$$

s_d = spread of component d over the round-1 states.

Cost-weighted

$$\mathcal{L} = \operatorname*{mean}_{n,j,d} \left(\frac{a_n\, g_{njd}\, r_{njd}}{D_{\text{ref}}}\right)^2, \qquad g_{nj} = (1,\ 1,\ \ell_{nj},\ \ell_{nj}),$$

- $\ell_{nj} = z_1 - (z_n + c_j \Delta z) + \Delta z$: the distance left to travel, plus one step. A slope error is multiplied by it.
- $D_n = \kappa\, |q/p|_n\, L \int |B|\, dz$: the total bend of track n over the region.
- D_{ref}: the median of D_n over the round-1 states.
- $a_n = \text{clamp}\big(\sqrt{W(p_n)}\, D_{\text{ref}}/D_n,\ [m/5,\ 5m]\big)$, m = batch median.
- $W(p) = 1$ inside the momentum window, Gaussian in $\log p$ outside (width $\ln 2$), floor 0.05. Windows: 10–50 GeV or 3–8 GeV.

7 Scores

On the test tracks, against the reference. Radial error $\sqrt{\Delta x^2 + \Delta y^2}$.

- **End of region**: chain the network N times from z_0; compare at z_1.
- **Single step**: one step from the reference state on each start plane; compare with the reference one step later.